CE 400/500

Special Topics: AI for Chemical Engineers

Lecture 9: Classification, Hands-on Practice on Tree-Based Models for Process Monitoring and Fault Detection

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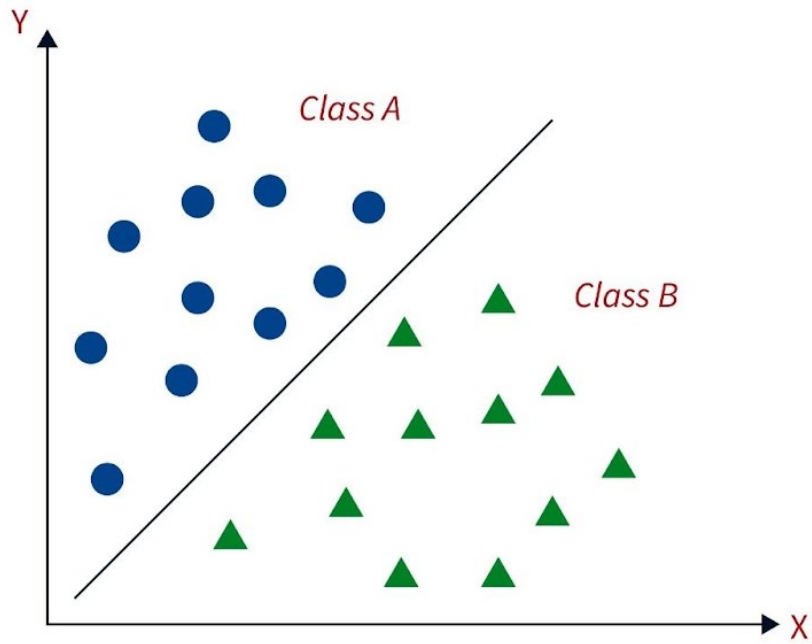
Classification

Classification is the task of predicting which of a set of classes (categories) an example belongs to. For example,

- predicting whether a given email is “spam” or “not spam”;
- predicting whether a chemical industrial process at any moment is “normal” or “not normal (faulty)”.



Confusion Matrix

There are four possible outcomes for each output from a binary classifier. For a fault detection classifier where **positive class is faulty process (category of greater interest) and negative class is normal process**, if you lay out the ground truth as columns and the model's prediction as rows, the following table, called a **confusion matrix**, is the result:

	Actual positive	Actual negative
Predicted positive	**True positive (TP):** A faulty process correctly classified as a faulty process.	**False positive (FP)**: A normal process misclassified as faulty process. These are the normal process at a moment that mistakenly detected as faulty.
Predicted negative	**False negative (FN)**: A faulty process misclassified as normal process. These are faulty process that are not caught by the detector.	**True negative (TN)**: A normal process correctly classified as normal.

https://developers.google.com/machine-learning/crash-course/classification/thresholding

Confusion Matrix (cont'd)

	Actual positive	Actual negative
Predicted positive	**True positive (TP):** A faulty process correctly classified as a faulty process.	**False positive (FP)**: A normal process misclassified as faulty process. These are the normal process at a moment that mistakenly detected as faulty.
Predicted negative	**False negative (FN)**: A faulty process misclassified as normal process. These are faulty process that are not caught by the detector.	**True negative (TN)**: A normal process correctly classified as normal.

- The total in each row gives all predicted positives (TP + FP) and all predicted negatives (FN + TN), regardless of validity.
- The total in each column, meanwhile, gives all real positives (TP + FN) and all real negatives (FP + TN) regardless of model classification.

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Evaluation Metrics for Classifier

True and false positives and negatives can be used to calculate several useful metrics for evaluating models. Which evaluation metrics are most meaningful depends on the specific model and the specific task, the cost of different misclassifications, and whether the dataset is balanced or imbalanced. Metrics to be covered include:

- Accuracy
- Recall
- False positive rates
- Precision
- F1 score (advanced)

Accuracy

Accuracy is the proportion of all classifications that were correct, whether positive or negative. It is mathematically defined as:

$$\text{Accuracy} = \frac{\text{correct classifications}}{\text{total classifications}} = \frac{TP + TN}{TP + TN + FP + FN}$$

- In the faulty detection example, accuracy measures the fraction of all process conditions correctly classified. A perfect model would have an accuracy of 1.0, or 100%.
- Because it incorporates all four outcomes from the confusion matrix, given a **balanced** dataset, with similar numbers of examples in both classes, accuracy can serve as a coarse-grained measure of model quality. For this reason, it is often the **default** evaluation metric.
- For heavily **imbalanced** datasets, where one class appears very rarely (e.g., normal process), accuracy would be useless (will come back to this in the hands-on practice).

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

Recall (or True Positive Rate)

The **true positive rate (TPR)**, or the proportion of all actual positives that were classified correctly as positives, is also known as **recall**. Recall is mathematically defined as:

$$\text{Recall (or TPR)} = \frac{\text{correctly classified actual positives}}{\text{all actual positives}} = \frac{TP}{TP + FN}$$

False negatives are actual positives that were misclassified as negatives, which is why they appear in the denominator.

A hypothetical perfect model would have a recall (TPR) of 1.0, or a 100% detection rate.

In the fault detection, **recall** measures the fraction of faulty process that were correctly classified as faulty. This is why another name for recall is **probability of detection.**

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

False Positive Rate

The **false positive rate (FPR)** is the proportion of all actual negatives that were misclassified as positives, also known as the **probability of false alarm**. It is mathematically defined as:

$$\text{FPR} = \frac{\text{incorrectly classified actual negatives}}{\text{all actual negatives}} = \frac{FP}{FP + TN}$$

A perfect model would have zero false positives and therefore a FPR of 0.0, which is to say, a 0% false alarm rate.

In the fault detection example, FPR measures the fraction of normal condition that were misclassified as faulty condition, or the model's **rate of false alarms**.

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

Precision

Precision is the proportion of all the model's positive classifications that are actually positive. It is mathematically defined as:

$$\text{Precision} = \frac{\text{correctly classified actual positives}}{\text{everything classified as positive}} = \frac{TP}{TP + FP}$$

A hypothetical perfect model would have zero false positives and therefore a precision of 1.0.

In the fault detection example, precision measures the fraction of conditions classified as faulty that were actually faulty.

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

Summary

	Actual positive	**Actual** negative
Predicted positive	True positive (TP)	False positive (FP)
Predicted negative	False negative (FN)	True negative (TN)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

Choice of Metric and Tradeoffs

Metric	Guidance
Accuracy	• Use as a rough indicator of model training progress/convergence for balanced datasets. • For model performance, use only in combination with other metrics. • Avoid for imbalanced datasets. Consider using another metric.
Recall (True positive rate)	Use when false negatives (e.g., misclassified faulty process) are more expensive than false positives (e.g., misclassified normal process).
False positive rate	Use when false positives are more expensive than false negatives.
Precision	Use when it's very important for positive predictions to be accurate.

The F1 score could be used if both precision and recall need to be considered.

https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall

F1 Score

The F1 score is the harmonic mean (a kind of average) of precision and recall. Mathematically, it is given by:

$$\mathrm{F1} = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}$$

This metric balances the importance of precision and recall, and is preferable to accuracy for **class-imbalanced** datasets.

When precision and recall both have perfect scores of 1.0, F1 will also have a perfect score of 1.0. More broadly, when precision and recall are close in value, F1 will be close to their value. When precision and recall are far apart, F1 will be similar to whichever metric is worse.

Hands-on: Classification with Decision Tree and Random Forest for Process Fault Detection

See Google Colab notebook